Summary of DDA3020 Machine Learning（2026 Spring）

Link to OCTE: https://octe.cuhk.edu.cn/a/login

QR code to OCTE:

Summary（2026 Spring）

Lecture 1 Introduction (45 pages)
1. Definition of ML (15)
2. Main branches of ML (20,21)
3. Supervised vs unsupervised learning (22)
4. Classification vs regression (22)
5. Basic concepts in supervised learning（37-40）

Lecture 2 Probability (32 pages)
1. Basic concepts of sample space, event, random variables, state space (4-6)
2. Different probabilities, Bayesian rule (8-12)
3. Expectation and variance of random variables (14,18)
4. Some widely used distributions, their formulations, expectations and variances (Bernoulli, Binomial, Gaussian) (20-22)
5. Definitions, formulations and properties of information, entropy, cross entropy, KL divergence (25-32)

Lecture 3 Linear Algebra (34 pages)
1. Vector, matrix, and their norms (5-17)
2. Matrix inverse, determinant, independence (19-23)
3. Three types of linear systems: Even-determined, Over-determined, Under-determined (25-34)

Lecture 4 Optimization (35 pages)
1. Affine set, Convex set (4,5)
2. Convex function and its properties (7,10,11,13)
3. Convex optimization, local/global optima (19,21)

4. Gradient descent (27)
5. Lagrangian function and KKT condition (31,32, memorize them)

Lecture 5 Linear regression (80 pages)

1. Definition (26)
2. Objective functions of two perspectives: Deterministic and Probabilistic (27-30)
3. Two solutions: analytical (33-34) and gradient descent (39-40)
4. Linear regression with multiple outputs (43-44)
5. Linear regression for classification (50)
6. Ridge regression: two motivations, their objective functions and solutions (57-62)
7. Polynomial regression: motivation, formulation, solution (63-69)
8. Lasso regression: motivation, formulation, solution (72-74)
9. Robust regression: motivation, formulation, solution (75-80)

Lecture 6 Logistic regression (47 pages)

1. Sigmoid function and decision boundary (15-17)
2. Cross entropy loss: formulation and property (23-27)
3. Gradient descent for logistic regression (28)
4. Multi-class classification, softmax regression, its gradient (34-35)
5. Regularized logistic regression: motivation, formulation and solution (40-41)
6. Probabilistic perspective of logistic regression (44-45)

Lecture 7 SVM I (65 pages)

1. Concept of large margin (12)
2. Derivation 1 of SVM formulation: large margin (19-23)
3. Derivation 2 of SVM formulation: hinge loss (26-29)
4. Lagrange duality and KKT conditions (31-32)
5. Derivation of the solution to SVM by Lagrange duality, and the solution interpretation (34-40)
6. SVM with slack variables: motivation, formulation, solution and solution interpretation (43-50)
7. SVM with kernels: motivation, formulation (53-55)
8. Multi-class SVM (60,61)
9. SVM vs. Logistic regression (62,63)

Lecture 8 Decision Tree & Random Forest (62 pages)

1. Motivation (4-5)
2. Definition (10) and Basic Terminologies (14)
3. Classification Trees (17-21)
4. Regression Trees: MSE (34,35)
5. Overfitting and Pruning (42,43)
6. Bagging (54-57) and Random forests (60,61)

Lecture 9 Neural Networks (52 pages)

1. Concepts: neuron model, activation function, perceptron model (9-11)
2. Multi-layer feedforward neural networks (16)
3. Backpropagation and computational graph (26-42)

Lecture 10 Convolutional Neural Networks (66 pages)

1. Convolution layer: filter, convolve (30), activation map (35), stride (43), padding (51)， summary (57)
2. Pooling layer (61-62)
3. Fully connected layer (65)

Lecture 11 Recurrent Neural Networks (60 pages)

1. Basic RNN architecture layer (12-21)
2. Training RNN (28-33)
3. LSTM (35)
4. Transformer (50, 53)

Lecture 12 Over/under-fitting, Bias-variance tradeoff (40 pages)

1. Overfitting, underfitting and model complexity (5, 9, 12)
2. Bias and Variance (21, 25)
3. Bias-variance tradeoff (27-31)

Lecture 13 Performance Evaluation (49 pages)

1. cross validation and Hyper-parameter tuning (10, 12-15)
2. Evaluation metrics for regression (20)
3. Evaluation metrics for classification: confusion matrix (23-24), accuracy/precision/recall (25), four rates (27, 32-34), ROC (36), AUC (37-44)

Lecture 14 Introduction to Unsupervised learning (28 pages)

1. Two motivations (5-6)
2. Main Approaches: you just need to know their tasks (11)

Lecture 15 K-means (43 pages)

1. Definition (5), basic algorithm (24)
2. Optimization perspective of K-means Clustering (26-32)
3. Evaluation of clustering (37-40)

Lecture 16 GMM-EM (58 pages)

1. Definition and formulation (3-4)
2. Maximum likelihood (10-13)
3. Latent Variable (25-27)
4. EM for fitting latent variable models (33-45)

Lecture 17 PCA (41 pages)

1. Projection onto a subspace (4-6)
2. Dimensionality Reduction (8-10)
3. PCA: motivation (13), maximal variance (15-16), minimal reconstruction error (18-19)
4. Algorithm (21-26)